

The recursive block decomposition of the CRT split

A proven torsion-localization, a per-block rank law with its residual now closed,
and a sofa-computable block count for Fermat cells

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Version 2.0 (eigencut) · 4 June 2026 (Version 1.0: 1 June 2026)

Abstract

This note is the companion to *The prime-power reduction frontier* (THE_FRONTIER_6_6.md). The Frontier explained why a fast cohomology reduction lies on composite degree—the cyclotomic polynomial factors over $\mathbb{F}_p$ —and its constructive dual, the **CRT block split**, turned that same factorization into a memory lever. Here we report the exact structure found inside that split. Three results are given in descending order of how firmly they stand: (i) **proven**—torsion localizes to a single CRT block, by centrality of the idempotents; (ii) a per-block rank law—each block’s rank is a product of two Degtyarev–Shimada recursion ladders, each a reduced-dimension Fermat rank—**whose single open residual of Version 1.0 is now closed** (§8): the *eigencut identity* proves the CRT idempotent cut and the Degtyarev–Shimada coordinate-zeroing cut are the same linear projection, in every characteristic; (iii) **verified byte-exact**—the number of blocks of a cell is computable in closed form from elementary cyclotomic number theory. Version 2.0 also corrects a section label carried from early campaign notes (the coordinate construction lives in §4.6 of the source, not a “§17”), and records, reading the complete source to its end, where the coupling of three or more pairs lives: by the source’s own equivalence (§4.6, an *if and only if*), the freedom of that coupling in recursive dimension $2s$ and Conjecture 1.2 in dimension $2s$ are one statement. The torsion counterexample (the *swan*) has not been found; every measured block is primitive. What the decomposition adds is an explicit recursive anatomy of these cells’ integral cohomology together with a proof that any torsion is confined to one computable block—the per-block elementary-divisor structure that Aljovin–Movasati–Villaflor explicitly left uncomputed.

How to read this. Section 1 fixes notation. Section 2 is a self-contained proof—torsion localizes to one block—and may be read alone. Section 3 is the main result: a per-block rank law, validated out of sample; its single open residual, stated in §3.2 as in Version 1.0, is **closed in §8**. Sections 4–5 give the block count and its conjectural extension. Section 6 is the honest scope and the graveyard of refuted approaches. Section 8 is the Version 2.0 addendum: the residual closed, the source section corrected, the coupling located. The punchline, for a reader in a hurry: localization is proven; the rank law’s value is proven in the companion theorem and its cut identity is now the eigencut theorem; the swan has not been found.

1 Notation and setting

Fix a Fermat cell (n, m) with $n = 2d$ even, and a prime $p \mid m$. Over $\mathbb{F}_p$ the polynomial $t^m - 1$ (equivalently the relevant φ) factors; grouping the $(t - 1)$ -primary part as A and the remainder as B , the per-variable quotient ring splits by the Chinese Remainder Theorem. Write $\dim A$ and

$\dim B$ for the $\mathbb{F}_p$ -dimensions of the two factors, with $\dim A + \dim B = m - 1$, and $NV = n + 1$ for the number of variables. The cohomology module splits into blocks indexed by which of the NV variables sit in A and which in B .

For a two-component cell ($A \times B$ —the case in which every measured cell of this campaign lives) there are 2^{NV} blocks, indexed by a binary mask of length NV ; the number of B -variables in a block is its **popcount** k . The general component count is given in Section 4.

Throughout, $DS_n(m)$ denotes the Degtyarev–Shimada closed-form rank of $L(X)$ for the Fermat cell of dimension n and degree m (Remark 4.4 of [1]). This is the *closing rank* $= \text{DIM} - \dim_{\mathbb{C}}$, equal to the quantity the prime-field reduction engine sums to; it is *not* $\dim_{\mathbb{C}}$ itself. All ranks below are in this sense.

2 Result one (proven): torsion localizes to a single block

Theorem 1 (Localization). *For every cell and every prime $p \mid m$, the integral cohomology module M decomposes as a direct sum over the CRT blocks. Consequently any torsion subgroup of M is confined to a single block—the unique block whose rank drops modulo p .*

Proof. The CRT idempotents e_i effecting the split are polynomials in the shift variable t : they are the standard idempotents $\prod_{j \neq i} (t - r_j)/(r_i - r_j)$ lifted from the factorization of the minimal polynomial. A polynomial in t commutes with multiplication by t ; since the module structure is generated by the shift, each e_i is central. The idempotents are orthogonal ($e_i e_j = 0$ for $i \neq j$) and complete ($\sum_i e_i = 1$)—the defining CRT relations. A central, orthogonal, complete system of idempotents yields a direct-sum decomposition $M = \bigoplus_i e_i M$ into submodules. A direct sum has a block-diagonal presentation matrix, and the Smith normal form of a block-diagonal matrix is the concatenation of the Smith normal forms of its blocks. Hence the elementary divisors—equivalently, the torsion—partition across the blocks, and any torsion is localized to the block(s) carrying it. $\square$

Significance. This is the rigorous half of a torsion detector. For every cell, with no computation and no unproven law, it says *where* a counterexample must be: in the one block that drops rank modulo p . It requires no dictionary between our blocks and any external construction, and no transfer of an external corollary; it follows directly from the idempotents being polynomials in the shift, which they are by construction.

What it does not give. Localization says *where*, not *what*. It does not by itself give the rank a primitive block should carry (that is Result Two), nor does it constitute a tested detector (see Section 6).

Verifier. `verifiers/CENTRALITY_DETECTOR_VERIFIER.py` constructs the idempotents for several cells (characteristic 2, 3, 5; $m = 6, 12, 15$) and checks byte-exact that they are central, orthogonal, idempotent, and complete—the hypotheses of Theorem 1.

3 Result two: the per-block rank is a product of two recursion ladders

Proposition 1 (Per-block product law). *Define the recursion ladder*

$$\text{rec}(d, \ell) = \begin{cases} 1 & \ell = 0, \\ d & \ell = 1, \\ DS_{2(\ell-1)}(d+1) & \ell \geq 2. \end{cases}$$

For a block with k of the NV variables in B (so $NV - k$ in A), set $\ell_A = \lceil (NV - k)/2 \rceil$ and $\ell_B = \lceil k/2 \rceil$. Then

$$\text{rank}(\text{block}) = \text{rec}(\dim A, \ell_A) \cdot \text{rec}(\dim B, \ell_B).$$

The block rank is a product of two ladders—one running the recursion on the A factor with the A -variables, the other on the B factor with the B -variables. Each ladder value $\text{DS}_{2(\ell-1)}(d+1)$ is the Degtyarev–Shimada rank of a Fermat cell of reduced dimension $2(\ell-1)$ and grade $d+1$: each block is, numerically, a smaller Fermat cell.

3.1 The evidence, in order of strength

Per-block byte-exact match. The law reproduces every individual block rank—not merely totals—on all four fully dumped cells: (4, 6) char 2 ($\dim A=1, \dim B=4$), (4, 10) char 2 ($\dim A=1, \dim B=8$), (4, 6) char 3 ($\dim A=2, \dim B=3$), and (4, 12) char 3 ($\dim A=2, \dim B=9$). For instance the (4, 6) char 3 block values 20, 18, 18, 38, 38, 141 (by popcount class) are each reproduced.

Blind out-of-sample prediction. The ladder multipliers were extracted from cells with $\dim A \in \{1, 2\}$ only. The law was then applied, unseen, to (4, 15) char 5, which has $\dim A = 4$ —a value never used in fitting. It predicted the block ranks by popcount as 400, 360, 360, 1080, 1080, 10900; the measured distinct values 360, 1080, 10900 match. A curve fit cannot extrapolate to a new $\dim A$; a product of fixed recursion ladders does, because the A -side multiplier is not a free per- $\dim A$ parameter—it is the same $\text{rec}(\cdot)$ evaluated at $\dim A$. This is the decisive evidence that the structure is real and not a fit.

A measured high-level monster. The cell (4, 14) char 2 ($\dim A=1, \dim B=12$) closed primitive on the 8 GB laptop at peak 2.218 GB, and its largest block (popcount 5, hence $\ell_B = 3$) measured $19,920 = \text{DS}_4(13)$ byte-exact—the first directly measured high- ℓ block at this $\dim B$, confirming the ladder value with measured data rather than deduction.

3.2 The single open residual of Version 1.0 — now closed (§8)

What was proven in Version 1.0 is the *form* (a product of two ladders), confirmed past fitting by the blind prediction. What was measured but not then proven is the cut identity: that the CRT idempotent projection (the $(t-1)$ -vs- $g(t)$ ring split) realizes the Degtyarev–Shimada coordinate-zeroing cut

$$X(2s) := W_s \cap \{z_{2s+2} = \cdots = z_{n+1} = 0\}$$

of [1, §4.6]—that our block and $X(2s)$ are the same reduced-dimension sub-Fermat, not merely two objects of equal rank. **This residual is closed in §8 by the eigencut identity**, and the ladder *value* is proven for every block dimension in the companion theorem [4]. (The construction was cited in Version 1.0 as “§17” of the source; the correct section is §4.6—see §8.1.)

Verifiers. `GENERAL_PRODUCT_LAW_VERIFIER.py` checks the product law per block on the four dumped cells and reproduces the blind (4, 15) char-5 prediction; `DS_CONNECTION_VERIFIER.py` checks the reduced-dimension/grade identification against the source construction; `TRIPLE_TIE_VERIFIER.py` checks the double-factorial fingerprint below.

3.3 The triple tie: one fingerprint, three faces

The double factorial $(2s+1)!! = 1, 3, 15, 105$ appears in three places that the recursion shows to be one:

1. the leading coefficient of DS_{2s} (Remark 4.4): $\text{DS}_2 \sim 3m^2$, $\text{DS}_4 \sim 15m^3$, $\text{DS}_6 \sim 105m^4$;

2. the subspace-count prefactor in [1, §4.6]: W_s contains $(2s+1)!! \cdot m^{d+1}$ linear subspaces;
3. the multiplicity backbone of the recursive-dimension blocks (the binomial palindrome below).

That the same double factorial governs all three is the fingerprint that the CRT block, the DS_{2s} value, and the count of s -subspaces in $X(2s)$ are the same object—corroboration that §8 now upgrades to the proven cut identity.

3.4 Separability and the binomial palindrome

Two structural facts fall out of the form, both verified.

Separability. The block rank depends on $(\dim B, c)$ only, *not* on the mother dimension n . Measured: $(6, 10)$ char 2 ($n=6, \dim B=8$) and $(4, 10)$ char 2 ($n=4, \dim B=8$) produce identical low-block values 1, 8, 8, 168, 168. The mother dimension sets only how many recursion rungs exist (up to recursive dimension n); the rung *values* are shared—exactly what the source tensor predicts, the low-dimension factor being independent of the total dimension.

Palindrome. Grouping blocks by recursion level, the multiplicities are palindromic: $n=4$ gives 1, 15, 15, 1; $n=6$ gives 1, 28, 70, 28, 1; $n=8$ gives 1, 45, 210, 210, 45, 1. Each sums to 2^{NV} and is symmetric under $c \leftrightarrow (\text{top} - c)$. The pairing observed from the start—adjacent levels carrying equal weight—is this binomial palindrome.

4 Result three (verified byte-exact): the block count is computable from the sofa

The $A \times B$ (two-component) picture is a special case. In general $t^m - 1$ factors over $\mathbb{F}_p$ into more than two coprime primary components, and the per-variable ring splits into that many factors.

Proposition 2 (Block-count law). *Let m' be the prime-to- p part of m . The number of coprime primary components per variable is*

$$1 + \sum_{\substack{d|m' \\ d>1}} \frac{\varphi(d)}{\text{ord}_d(p)},$$

where φ is Euler's totient and $\text{ord}_d(p)$ is the multiplicative order of p modulo d . The total number of blocks of a cell is this count raised to the power NV .

The “1” is the $(t-1)$ -primary part; each divisor $d > 1$ of m' contributes $\varphi(d)/\text{ord}_d(p)$ irreducible factors of the d -th cyclotomic polynomial, each a primary component.

Evidence. Verified byte-exact against direct factorization of $t^m - 1$ over $\mathbb{F}_p$ for all $p \in \{2, 3, 5, 7, 11\}$ and $m \in [3, 24]$ with $p \mid m$ —28 cases, zero mismatches. The count is frequently more than two: $(4, 14)$ char 2 $\rightarrow 3$; $(4, 18)$ char 2 $\rightarrow 3$; $(4, 12)$ char 3 $\rightarrow 3$; $(4, 24)$ char 3 $\rightarrow 5$; $(4, 20)$ char 5 $\rightarrow 4$; $(4, 21)$ char 7 $\rightarrow 3$.

Consequence for the hunt. With Theorem 1 (torsion in one block), this gives from the armchair the size of the swan's hiding space per cell—how many blocks the counterexample could occupy—before any computation. $(4, 14)$ char 2 has $3^5 = 243$ rooms; $(4, 6)$ char 2 has $2^5 = 32$. Richer prime structure means a finer lattice with more, smaller rooms.

A correction this carries. The “ 2^{NV} masks” framing used while the decomposition was first mapped is the two-component special case. Every cell measured in this campaign is two-component, so the per-block tables and the validated law of Section 3 stand exactly as stated; but the general detector must use the count above. The > 2 -component law is the conjecture of Section 5.

Verifier. `BLOCK_COUNT_LAW_VERIFIER.py`.

5 The r-ladder: a flagged conjecture for more than two components

Conjecture 1 (r-ladder). *For a cell whose per-variable ring has r coprime primary components $C_1, \dots, C_r$, a block (indexed by the distribution $n_1 + \dots + n_r = NV$ of the variables among the components) has rank $\prod_{i=1}^r \text{rec}(\dim_i, \lceil n_i/2 \rceil)$ —a product of r ladders, one per component.*

This is a conjecture and is flagged as such. There is no measured > 2 -component cell in hand to test it, and it carries a free ambiguity only data can resolve: the ladder’s dimension parameter could be the component’s $\mathbb{F}_p$ vector-space dimension or its Fermat-dim A -type value (these differ—for $(4, 14)$ char 2 the $(t - 1)$ -component has vector dimension 2 but Fermat type $\dim A = 1$). The localization of Theorem 1 holds for all components (its proof does not assume two); what is unverified is the per-block value law for $r > 2$.

The datum that would test it. The smallest virgin > 2 -component cell that is RAM-reachable on 8 GB is $(4, 14)$ char 2 split into its true three components (the $(t - 1)$ part and the two cubic factors of $t^7 - 1$ over $\mathbb{F}_2$). The current dump engine groups the two cubics into a single B block (running the cell as two-component), so the existing $(4, 14)$ run does not test the conjecture; an engine that splits B into its two cubic blocks (a $3^5 = 243$ -block lattice) is required. This is logged as the next measured target.

6 Honest scope

- **Proven:** torsion localizes to a single CRT block, for every cell (Theorem 1).
- **The per-block product law (Proposition 1):** the form confirmed out of sample (blind prediction); the value proven for every block dimension in the companion theorem [4]; the cut identity — the residual of Version 1.0 — proven in §8 (the eigencut identity).
- **Verified byte-exact:** the sofa block-count law (Proposition 2).
- **Conjectural, untested:** the r-ladder generalization to > 2 -component cells (Section 5).

Detector status. Localization is the *where*; the value law is the *what*. With the residual closed (§8) and the value proven [4], a measured departure from $\text{DS}_{2(\ell-1)}(d + 1)$ in a recursive-dimension- ≥ 4 block would be the deviation event of §8.3. The standing rule remains binding: before any future nonzero scar is trusted as a real torsion verdict, a synthetic positive control—a fabricated system with known torsion—must be shown to make the detector fire correctly.

The graveyard (dead veins, recorded so they are not retried). *The partition-mask dictionary.* An early plan to certify the law by mapping the Degtyarev–Shimada coordinate-partitions J (count $(2d + 1)!!$ —15 for $n=4$, 105 for $n=6$) to our CRT masks (count 2^{NV} —32 for $n=4$, 128 for $n=6$) piece by piece was refuted with data: the counts misalign ($15 \neq 32$, $105 \neq 128$) and our linear layer has no Degtyarev–Shimada counterpart. The dictionary does not exist as a piecewise correspondence; the correct road to certification is the centrality proof of Theorem 1, which needs no dictionary. *The “total = DS” check is vacuous as evidence.* Summing the product law over all masks reproduces $\text{DS}_n(m)$ for any split $\dim A + \dim B = m - 1$, including non-existent splits—a binomial identity true by construction. It proves nothing about per-block correctness. It is a necessary consistency check (it reflects the characteristic-independence of $\dim_{\mathbb{C}}$), but it is not evidence for the block structure; the real evidence is the per-block match and the blind prediction.

What this is not. This concerns the integral Hodge conjecture for Fermat varieties—a specific, bounded problem. It is not the Clay Millennium (rational) Hodge conjecture and makes no claim

on it. The swan has not been found; every measured block is primitive. What the decomposition adds is an explicit recursive anatomy of these cells’ cohomology, a proof that any torsion is confined to one computable block, and a from-the-sofa map of where to look and what number would betray a counterexample—the per-block elementary-divisor structure that Aljovin–Movasati–Villaflor [2] explicitly left uncomputed.

7 Reproducibility

Each result has a self-contained Python verifier that recomputes the Degtyarev–Shimada values from scratch:

```
python3 verifiers/CENTRALITY_DETECTOR_VERIFIER.py # Thm 1: idempotents central
python3 verifiers/GENERAL_PRODUCT_LAW_VERIFIER.py # Prop 1: product law + blind dimA=4
python3 verifiers/DS_CONNECTION_VERIFIER.py       # Prop 1: reduced-dim/grade vs source
python3 verifiers/TRIPLE_TIE_VERIFIER.py          # the double-factorial fingerprint
python3 verifiers/BLOCK_COUNT_LAW_VERIFIER.py      # Prop 2: block count vs factorization
python3 verifiers/EIGENCUT_IDENTITY_VERIFIER.py   # Sec 8: the eigencut identity, 10 cells
python3 verifiers/HINGE1_DEGREE_BOUND_VERIFIER.py # Sec 8: the degree bound, m = 4..15
python3 verifiers/RIFLE_FINAL.py 4 35             # the sofa predictor, [FIRM]/[CONJ]
```

RIFLE_FINAL.py is the from-the-sofa predictor: it states a primitivity verdict and $\dim_{\mathbb{F}_p}$ for any cell (n, m) , $n \in \{4, 6\}$, for every prime $p \mid m$, flagging each prime [FIRM] (two-component, the validated law applies) or [CONJ] (more than two components, resting on the unverified r-ladder). The flag is the discipline: a conjectural prediction is never printed as a confirmed one.

8 Addendum (Version 2.0, 4 June 2026): the residual closed, the source section corrected, the coupling located

This addendum records three facts settled after Version 1.0 was sealed. The body above is carried unaltered except where explicitly marked; where a statement below supersedes one above, the superseded line is named.

8.1 The source-section correction

The construction cited in Version 1.0 as “§17” of the source—the explicit sub-Fermat obtained by zeroing coordinates, $X(2s) := W_s \cap \{z_{2s+2} = \cdots = z_{n+1} = 0\}$, with its $(2s+1)!! \cdot m^{d+1}$ linear subspaces—lives in §4.6 of Degtyarev–Shimada (the paper has five sections; the tensor decomposition is the proof of their Corollary 1.7). The “§17” was a label error for that same object, carried from early campaign notes and settled against the complete source text. Every mathematical statement referring to it is unaffected; only the label was wrong. The body above has been relabelled accordingly.

8.2 The residual of §3.2 is closed — the eigencut identity

Version 1.0’s single open residual was: *prove that the campaign’s CRT cut is the Degtyarev–Shimada coordinate cut—the same object, not merely the same count*. That residual is now **proven**. On the per-variable ambient $V = k[t]/(t^m - 1)$, the generalized eigenspace $E_1(t) = \ker((t - 1)^m)$ —the part fixed by the Galois rotation γ_j , i.e. what the coordinate cut $z_j = 0$ retains—equals $\text{im}(e_A)$, the

image of the CRT idempotent, in every characteristic; the two cuts are the *same linear projection*. The proof is the primary decomposition of V as a $k[t]$ -module, in the same elementary register as the campaign’s red link, with its one load-bearing hypothesis—the Fermat degree bound $0 \leq \nu < m$ of Degtyarev–Shimada Lemma 4.1, under which “exponent $\equiv 0 \pmod{m}$ ” collapses to “exponent 0”—stated and verified ($m = 4..15$) rather than assumed. Anchored byte-exact in ten cells across characteristics $\{0, 2, 3, 5\}$ and $\dim A \in \{1, 2, 3, 4, 5, 8\}$, including the maximal towers $(8, 2)$ and $(9, 3)$. Full statement, proof and verifiers: `THE_EIGENCUT_IDENTITY.md`, `EIGENCUT_IDENTITY_VERIFIER.py`, `HINGE1_DEGREE_BOUND_VERIFIER.py`.

Consequently, Result Two’s Version 1.0 status line (“strongly validated, one open residual”) is superseded: the form was confirmed past fitting by the blind $(4, 15)$ prediction, the value is proven for every block dimension in the companion theorem [4] (the red link), and the cut identity is the eigencut theorem. The corresponding summary lines in §3.2 and §6 are updated above.

8.3 The coupling, located in the source

Reading the complete Degtyarev–Shimada text to the end settles where the next object beyond this document’s ladder lives. Their Theorem 1.1(d), verbatim: the coupled module is $\bar{C}_K := (\bigoplus_{J \in K} \bar{R}_J) / \bar{M}$, with $\bar{M}$ generated by the *single element* $\sum_{J \in K} 1_J$; each leg $\bar{R}_J$ is $\mathbb{Z}$ -free by their Lemma 4.1 (τ_J is a θ over the k -variables, no restriction on the number of variables). Their §4.5 proves the four torsion descriptions of Theorem 1.1 isomorphic; the freedom of the quotient by $\sum 1_J$ is addressed in their §4.6, which states, verbatim:

“Thus, this module is free (as an abelian group) if and only if so is $\bar{C}_{J(2s)}(2s)$, i.e., if and only if Conjecture 1.2 holds for Fermat varieties of dimension $2s$ in $\mathbb{P}^{2s+1}$.”

By that equivalence—the source’s own, an *if and only if*—the freedom of the coupling of three or more pairs in recursive dimension $2s$ and Conjecture 1.2 in dimension $2s$ are **one statement**; the eigencut identity (§8.2) makes the application to the campaign’s objects exact. Two legs sit below the equivalence (the within-pair closure is $\mathbb{Z}[t]/\varphi$, free by the red link’s $x \cdot c_0 = -1$; dimensions 0 and 2 are torsion-free by the source’s own results); three or more legs sit on it. Every primitivity verdict of the campaign is accordingly an instance of Conjecture 1.2 confirmed, and a deviation in a recursive-dimension ≥ 4 block, should one ever be measured, and a failure of Conjecture 1.2 in that dimension are the same event. The full narration is in the companion theorem [4], §8.

References

- [1] A. Degtyarev, I. Shimada, *On the topology of projective subspaces in complex Fermat varieties*. J. Math. Soc. Japan **68**:3 (2016), 975–996. doi:10.2969/jmsj/06830975. arXiv:1405.4683. (The criterion; §4.6, the coordinate construction $X(2s)$ and the tensor of Cor. 1.7; Remark 4.4, the closed-form rank.)
- [2] E. Aljovin, H. Movasati, R. Villafior, *Integral Hodge conjecture for Fermat varieties*. J. Symbolic Computation **95** (2019), 177–184. arXiv:1711.02628. (Tabulates the primitive lattice but leaves its elementary divisors uncomputed.)
- [3] R. Amichis Luengo, *The eigencut identity — the CRT idempotent cut is the Degtyarev–Shimada coordinate cut*. Campaign note, version 1.0, 3 June 2026. `THE_EIGENCUT_IDENTITY.md`.
- [4] R. Amichis Luengo, *The block-rank theorem for Fermat CRT blocks, and the red link that makes it unconditional*. Version 3.0 (closed, eigencut), 4 June 2026. `THE_NAIL_THEOREM.md` / `THE_NAIL_THEOREM_CLOSED_EIGENCUT.pdf`.